

When a Ride is the Missing Treatment

An investigation into transportation barriers and patient journeys within a regional health system.

TEAM
TEAM 13

INSTITUTION
Wesleyan University

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DATAFEST 2026

"Do patients who report a transportation barrier use acute care differently than otherwise similar patients?"



The Question: Does one transportation answer predict a different patient journey?

THE PREMISE

- **Missed Rides,** Transportation barriers often lead to missed
Downstream Costs: routine appointments.
- **Reactive** Without preventive access, patients may rely more heavily
Care: on acute settings like the ED.

"If transportation keeps people from care, I expect to see more reactive use of the hospital system—not as a choice, but as a consequence of blocked access."

WHAT I MEAN BY ONE TRANSPORTATION ANSWER

Transportation Needs (Q1)

"In the past 12 months, has lack of transportation kept you from medical appointments, meetings, work, or from getting things needed for daily living?"

Transportation Needs (Q2)

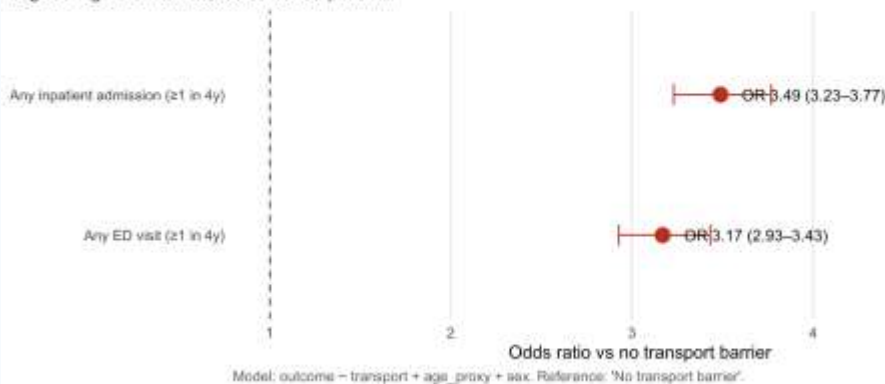
"In the past 12 months, have you ever had to worry about not having a way to get to your medical appointments?"

Transport barrier predicts acute-care use after age and sex adjustment

PRIMARY INFERENCE: ADJUSTED FOREST PLOT

Transport barrier raises odds of acute care ~3× (age- and sex-adjusted)

Logistic regression on 58,639 screened patients.



Logistic: outcome ~ transport + age + sex
n = 58,639 | Screening Rate ~6%

Any ED Visit

3.17

[95% CI: 2.93–3.43]

Any Inpatient

3.49

[95% CI: 3.23–3.77]

Barrier group is younger (median 51 vs 61) — effect is independent of age.

180-day ED return after index encounter (2022–2025)

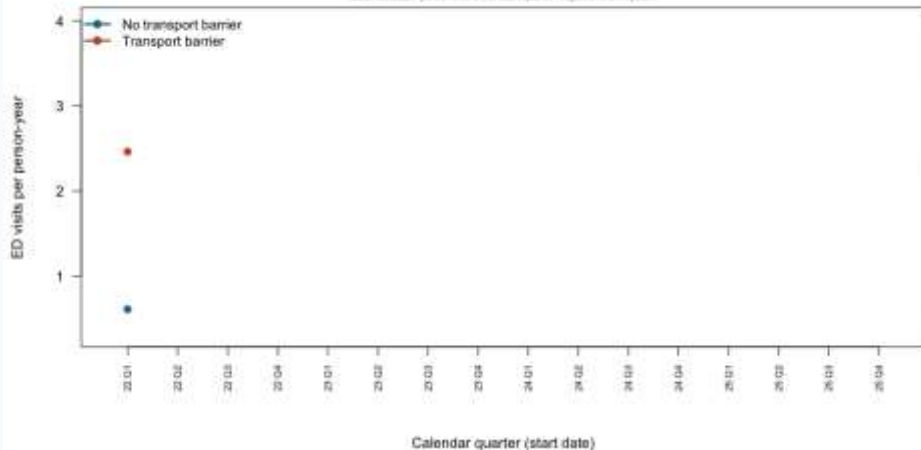
CONDITION (INDEX)	NO BARRIER — % WITH ED VISIT (N)	BARRIER — % WITH ED VISIT (N)
Hypertension (I10)	15% (18,890)	33% (672)
Type 2 Diabetes (E11)	17% (7,989)	40% (541)
CKD (N18)	21% (4,118)	37% (238)
AFib / Flutter (I48)	24% (4,166)	34% (183)

* Index = first qualifying encounter; barrier n smaller by cell—interpret as descriptive shares.

From Signal to Action: Moving Towards Solutions

ED visit intensity by calendar quarter (screened cohort)

Dataset 238 - perspective: all of each quarter - regional health system



STRATEGIC INTERVENTIONS

-  **Targeted Transit:** Rides and shuttles for high-risk acute-care users.
-  **Telehealth Bridge:** Virtual follow-ups to bypass physical distance.
-  **Pharmacy Delivery:** Home access for chronic disease management.



CRITICAL CAVEATS

-  **Selection Bias:** Non-random screening (~6% of population).
-  **Data Integrity:** ~20% of encounters lack matched diagnosis keys.
-  **Geographic Gaps:** Missing FIPS data limits spatial modeling.



One transportation screen identifies patients with ~3x adjusted odds of acute care. Ask the question more, and resource the response.